

SOCIAL SCIENCE

The fundamental advantages of temporal networks

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Most networked systems of scientific interest are characterized by temporal links, meaning the network's structure changes over time. Link temporality has been shown to hinder many dynamical processes, from information spreading to accessibility, by disrupting network paths. Considering the ubiquity of temporal networks in nature, we ask: Are there any advantages of the networks' temporality? We use an analytical framework to show that temporal networks can, compared to their static counterparts, reach controllability faster, demand orders of magnitude less control energy, and have control trajectories, that are considerably more compact than those characterizing static networks. Thus, temporality ensures a degree of flexibility that would be unattainable in static networks, enhancing our ability to control them.

Traditionally, network science has focused on static networks, whose links offer permanent connections between their nodes. Yet, it is increasingly recognized that most natural and social systems are best described as “temporal” networks, with links existing only intermittently. For example, within metabolic networks (1), the links correspond to brief chemical reactions; in social networks (2, 3), friendship links are inferred from face-to-face or digital communications of short duration. Such temporality has profound effects on most dynamical processes taking place on networks (4–7), slowing down synchronization and the diffusion of innovative information (4), impeding exploration and navigation (7), and raising barriers to accessibility (5). Similar limitations are expected for control—the ability to drive a system with input signals to any desired final state in finite time. Control is essential for the operation of most real systems (8–14), yet controllability normally requires the existence of continuous paths capable of carrying the input signals to the rest of the network (15–18). Whereas in static networks such signal-carrying paths are permanently available, in temporal systems, complete instantaneous paths between the inputs and the rest of the nodes are not guaranteed, potentially degrading our ability to control a system.

A temporal network is an ordered sequence of $m = 1, \dots, M$ separate networks on the same set of N nodes (Fig. 1, A and B), with each such “snapshot” m characterized by a (weighted) adjacency matrix $\mathbf{A}_m$ for a duration Δt_m . As we aim to uncover the role of the changing network

topology, rather than the effect of specific dynamics, we consider that, in each snapshot, the system is governed by the canonical linear time-invariant dynamics

$$\dot{\mathbf{x}}(t) = \mathbf{A}_m \mathbf{x}(t) + \mathbf{B}_m \mathbf{u}_m(t) \quad (1)$$

valid over the time interval $t \in [t_{m-1}, t_{m-1} + \Delta t_m]$,

where the $t_{m-1} = \sum_{j=1}^{m-1} \Delta t_j$ is the “switching time”

between snapshots $m - 1$ and m . We assume that the system spends a finite amount of time in each snapshot (1, 19), hence the Zeno phenomenon, or infinitely fast switching, is unlikely to occur in real temporal networks. The state vector $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_N(t)]^T \in \mathbb{R}^N$ captures the state of the whole system at time t , and $x_i(t)$ represents the state of node i , like the concentration of metabolite i within a cell. The input matrix $\mathbf{B}_m$ identifies the set of driver nodes through which we attempt to control the system using $p(\leq N)$ independent control inputs $\mathbf{u}_m(t) \in \mathbb{R}^p$, like manipulating the input concentration of p metabolites. To avoid conferring an unfair advantage to temporal networks, we use the same set of driver nodes across all snapshots, i.e., $\mathbf{B}_m = \mathbf{B}$ (Fig. 1C), and we assume that we have no control over the order of the snapshots nor over the timings of the topology changes, hence our influence on the system is confined to the control inputs (15–17, 20, 21). Consequently, in this work we can regard temporal networks with dynamics described by Eq. 1 as switched linear systems (22) with specified switching sequences. If the switching sequence depends on the state vector $\mathbf{x}(t)$, the system can display strong nonlinear behavior, even if the dynamics (Eq. 1) within each snapshot is linear (22).

To understand why temporal networks are easier to control than static networks, consider the static three-node network of Fig. 1D, which is uncontrollable by any single driver node. Indeed, we can show that for almost all values of the link weights, we can only steer the static network within a two-dimensional subspace of the three-dimensional state space (Fig. 1D). Yet, we can

also show that the temporal version of the same network—in which the two links are non-simultaneously active—is controllable by the top node (Fig. 1F). This is because although each snapshot of the temporal network is by itself an uncontrollable system, they contribute independent two-dimensional controllable subspaces (Fig. 1F), meaning that by combining the two, we can reach any point within the full three-dimensional state space. This renders the system as a whole controllable.

Consider a system initially at $\mathbf{x}_0 = \mathbf{0}$. By considering all possible trajectories from the initial time $t_0 = 0$ to the final time $t_f = t_M$, we can write the controllable space [see supplementary materials (SM) section S1.1] as

$$\Omega = \langle \mathbf{A}_M | \mathbf{B} \rangle + \sum_{m=1}^{M-1} \prod_{j=m}^{M-1} e^{\mathbf{A}_j \Delta t_j} \langle \mathbf{A}_m | \mathbf{B} \rangle \quad (2)$$

Here, $\langle \mathbf{A}_m | \mathbf{B} \rangle = \sum_{i=0}^{N-1} \mathbf{A}_m^i \mathbf{R}(\mathbf{B})$ denotes the controllable space of snapshot m , where $\mathbf{R}(\mathbf{B}) = \{\mathbf{B}\mathbf{v} | \mathbf{v} \in \mathbb{R}^p\}$ is the column space of $\mathbf{B}$. Hence, a temporal network of N nodes is controllable if and only if

$$\Omega = \mathbb{R}^N \quad (3)$$

meaning that we can steer the system to an arbitrary state of the state space $\mathbb{R}^N$ in finite time. For a static network, whose snapshots are identical (i.e., $\mathbf{A}_m = \mathbf{A}$), Eq. 3 reduces to the classic Kalman's rank condition for controllability (8).

According to Eq. 2, the controllable space will never shrink as the network structure changes, allowing us to determine the minimum number of snapshots S_t that must elapse before a temporal network becomes fully controllable. For comparison, we also calculate the minimum number of snapshots S_s for the corresponding controllable static network, which represents an aggregation of a temporal sequence of snapshots (SM section S4). Consider for example Fig. 1C, which shows a temporal network with four snapshots, none of which is individually controllable by using the top node as the sole driver node. According to Eq. 3, the temporal sequence becomes controllable at the second snapshot, i.e., $S_t = 2$ (Fig. 1F). By contrast, we must aggregate $S_s = 3$ snapshots to obtain a controllable static network (Fig. 1E). The difference between S_s and S_t captures the relative control benefits of a dense (but fixed) network topology versus a relatively sparse (but time-varying) topology, respectively. Although there is no theoretical guarantee that S_t is always less than S_s (fig. S1), as we show next, we find that real temporal networks reach controllability much faster than their static counterparts. Here “faster” refers to the number of snapshots we need to reach full controllability. Hence, the time to control (embodied in S_t and S_s) is distinct from t_f , representing the time a system needs to reach its final state (8, 15, 22–24).

To demonstrate the practical relevance of our finding, we explore an empirical communication data set collected by the SocioPatterns

社会科学

时序网络的根本优势

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大多数具有科学意义的网络系统都由时序链接特征，这意味着网络结构会随时间变化。链接时序性已被证明会通过破坏网络路径来阻碍许多动态过程，从信息传播到可访问性。考虑到时序网络在自然界中的普遍性，我们提出：网络的时序性是否存在任何优势？我们使用分析框架表明，与静态对应物相比，时序网络可以更快地达到可控性，所需控制能量低几个数量级，并且其控制轨迹比静态网络的特征轨迹要紧凑得多。因此，时序性确保了静态网络无法达到的一定程度的灵活性，增强了我们控制它们的能力。

传统上，网络科学主要关注静态网络，其

连接为节点之间提供永久性连接。然而，人们越来越认识到，大多数自然和社会系统最好被描述为“时序”网络，连接仅短暂存在。例如，在代谢网络（1）中，连接对应于短暂的化学反应；在社会网络（2, 3）中，友谊连接是根据面对面或数字通信的短暂时长推断出来的。这种时序性对网络上的大多数动态过程（4–7）产生深远影响，减缓同步和创新信息的扩散

（4），阻碍探索和导航（7），并提高可访问性的障碍（5）。类似的局限性也适用于控制——即通过输入信号在有限时间内将系统驱动到任何所需最终状态的能力。控制对于大多数实际系统的运行（8–14）至关重要，但可控性通常需要存在能够承载输入信号到达网络其他部分的连续路径（15–18）。在静态网络中，这种承载信号的路径是永久可用的，但在时序系统中，输入与其他节点之间的瞬时完整路径无法得到保证，可能会降低我们控制系统的能力。

时序网络是一个有序的 $m = 1, \dots, M$ 独立网络序列，这些网络位于相同的 N 节点集合上（图1, A和B），每个这样的“快照” m 都由一个（加权的）邻接矩阵 $\mathbf{A}_m$ 在持续时长 Δt_m 上表征。随着我们旨在揭示变化网络拓扑的作用，而非特定动力学的影响，我们认为在每个快照中，系统都受规范性线性时不变动力学支配

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拓扑控制，而非具体动力学效应，我们考虑，在每个快照中，系统都受规范性线性时不变动力学支配

$$\dot{\mathbf{x}}(t) = \mathbf{A}_m \mathbf{x}(t) + \mathbf{B}_m \mathbf{u}_m(t) \quad (1)$$

在时间间隔 $t \in [t_{m-1}, t_{m-1} + \Delta t_m]$ 内有效，其中 $t_{m-1} = \sum_{j=1}^{m-1} \Delta t_j$ 是“切换时间” $j=1$

在快照 $m - 1$ 和 m 之间。我们假设系统在每个快照中花费有限的时间（1, 19），因此，在真实的时序网络中，Zeno现象或无限快的切换不太可能发生。状态向量 $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_N(t)]^T \in \mathbb{R}^N$ 捕捉了系统在时间 t 的状态，而 $x_i(t)$ 表示节点 i 的状态，例如细胞内代谢物 i 的浓度。输入矩阵 $\mathbf{B}_m$ 通过我们试图使用 $p(\leq N)$ 个独立的控制输入 $\mathbf{u}_m(t) \in \mathbb{R}^p$ 来控制系统的驱动节点集，例如操纵 p 种代谢物的输入浓度。为了避免对时序网络给予不公平的优势，我们在所有快照中使用相同的驱动节点集，即 $\mathbf{B}_m = \mathbf{B}$ （图1C），并且我们假设我们对快照的顺序或拓扑变化的时机没有控制权，因此我们对系统的影响仅限于控制输入（15–17, 20, 21）。因此，在这项工作中，我们可以将动力学由方程1描述的时序网络视为具有指定切换序列的切换线性系统（22）。如果切换序列取决于状态向量 $\mathbf{x}(t)$ ，即使每个快照内的动力学（方程1）是线性的，系统也可以表现出强烈的非线性行为（22）。

要理解为什么时序网络比静态网络更容易控制，可以考虑图1D中的静态三节点网络，该网络无法被任何单个驱动节点控制。实际上，我们可以证明，对于几乎所有链路权重的值，我们只能将静态网络引导到三维状态空间中的二维子空间内（图1D）。然而，我们也可以

证明，该网络的时序版本——其中两条链路不会同时激活——可以被顶节点控制（图1F）。这是因为尽管时序网络的每个快照本身都是一个不可控系统，但它们各自贡献了独立的二维可控子空间（图1F），这意味着通过结合两者，我们可以到达三维状态空间内的任何点。这使得整个系统成为可控的。

考虑一个系统初始处于 $\mathbf{x}_0 = \mathbf{0}$ 。通过考虑从初始时间 $t_0 = 0$ 到最终时间 $t_f = t_M$ 的所有可能轨迹，我们可以写出可控空间 Ω ，详见补充材料（SM）部分 S1。1] 作为

$$\Omega = \langle \mathbf{A}_M | \mathbf{B} \rangle + \sum_{m=1}^{M-1} \prod_{j=m}^{M-1} e^{\mathbf{A}_j \Delta t_j} \langle \mathbf{A}_m | \mathbf{B} \rangle \quad (2)$$

Here, $\langle \mathbf{A}_m | \mathbf{B} \rangle = \sum_{i=0}^{N-1} \mathbf{A}_m^i \mathbf{R}(\mathbf{B})$ denotes the controllable space, where $\mathbf{R}(\mathbf{B}) = \{\mathbf{B}\mathbf{v} | \mathbf{v} \in \mathbb{R}^p\}$ is the column space of $\mathbf{B}$. Hence, a temporal network of N nodes is controllable if and only if

$$\Omega = \mathbb{R}^N \quad (3)$$

这意味着我们可以在有限时间内将系统引导至状态空间 $\mathbb{R}^N$ 的任意状态。对于静态网络，其快照是相同的（即 $\mathbf{A}_m = \mathbf{A}$ ），则式3简化为经典的可控性秩条件（8）。

根据式2，随着网络结构的变化，可控空间永远不会缩小，这使我们能够确定在时序网络完全可控之前必须经过的最少快照数 S_t 。为了比较，我们还计算了对应可控静态网络的最少快照数 S_s ，它表示一个时序快照序列的聚合（SM部分S4）。例如，图1C显示了一个包含四个快照的时序网络，其中没有任何一个快照可以通过仅使用顶部节点作为驱动节点而单独可控。根据式3，时序序列在第二个快照时变得可控，即 $S_t = 2$ （图1F）。相比之下，我们必须聚合 $S_s = 3$ 个快照才能得到一个可控的静态网络（图1E）。 S_s 与 S_t 之间的差异捕捉了密集（但固定）网络拓扑与相对稀疏（但时变）网络拓扑的相对控制优势。尽管没有理论保证 S_t 总是小于 S_s （图S1），但正如我们接下来所展示的，真实时序网络比其静态对应物更快达到可控性。这里“更快”指的是达到完全可控性所需的快照数。因此，控制时间（体现在 S_t 和 S_s 中）与 t_f 不同，后者表示系统达到最终状态所需的时间（8, 15, 22–24）。

为展示我们发现的实际意义，我们探索了SocioPatterns收集的一个实证通信数据集

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collaboration, capturing face-to-face conversations between the attendees of a conference (25); an ecological network, capturing antenna-body interactions between ants (26); a biological network with time-varying protein-protein binding interactions (27); and the real-time communication patterns observed in a technological network, extracted from observed data-packet exchanges within a mobile ad hoc network that emulates the true stationary communication activity of several mobile devices. For each data set, we condense interactions into snapshots over successive time windows of equal length Δt , a parameter that offers a measure of temporality: For small Δt , we obtain a large number of sparse, disjointed snapshots, whereas for large Δt , we approach a single snapshot corresponding to the full static network (Fig. 1, A and B). In this view, each snapshot represents the network structure over the full aggregation window Δt for the purposes

of the dynamics (Eq. 1). Figure 2 shows S_t and S_s for different time windows Δt , indicating that the number of snapshots required to control a temporal network S_t is less than that required for its static counterpart, S_s . The smaller the Δt , the more pronounced the advantage of temporal networks. For example, for the face-to-face interactions at $\Delta t = 10^3$, the temporal networks reach controllability after only 71 time steps (19.72 hours), whereas we must aggregate 185 snapshots over two days to obtain a static network that is controllable. The difference between S_t and S_s is significant in all systems, obtaining $P < 0.001$ for face-to-face interactions, $P < 0.05$ for ants, $P < 0.001$ for the protein network, and $P < 0.001$ for the technological network.

To determine to what extent these gains in controllability depend on the topology of the underlying snapshots, we calculate S_t and S_s for randomized versions of the empirical data, using

null models that permute the overall network structure, the relative time orderings of the edges, or both (19). In all cases, we find that $S_t < S_s$ (fig. S8), indicating that the observed advantage is independent of how the network changes in time. We also find that with increasing numbers of snapshots, temporal networks have more opportunities to reach full controllability than their static counterparts (fig. S9), a finding further validated by using a toy model that considers random switching sequences (SM section S4). Finally, we also explore whether the gains in controllability depend on the timing of the interactions between the nodes, specifically the interevent characteristics. For this, we randomly distribute time stamps for each interaction event (fig. S7), erasing the inherent burstiness (periods of intense activity separated by periods of relative quiescence) and other temporal correlations from the data (28), while leaving the number of

合作, 捕捉会议与会者之间的面对面交谈 (25); 一个生态网络, 捕捉蚂蚁之间的天线-身体互动 (26); 一个具有时变蛋白质-蛋白质结合相互作用的生物网络 (27); 以及在一个技术网络中观察到的实时通信模式, 这些模式从模拟多个移动设备真实静态通信活动的移动自组网络中的观测数据包交换中提取。对于每个数据集, 我们将交互浓缩到连续的时间窗口 Δt 内的快照中, Δt 是一个提供时间度量的参数: 对于小的 Δt , 我们获得大量稀疏、不连续的快照, 而对于大的 Δt , 我们接近一个对应于完整静态网络的单一快照 (图1, A 和B)。在这种观点下, 每个快照代表在完整聚合窗口 Δt 内的网络结构, 以供

动力学 (公式1)。图2显示了不同时间窗口 Δt 下的 S_t 和 S_s , 表明控制时序网络 S_t 所需的快照数量少于其静态对应物 S_s 。 Δt 越小, 时序网络的优势越明显。例如, 对于面对面互动 $\Delta t = 10^3$, 时序网络在仅71个时间步 (19.72小时) 后即可达到可控性, 而我们必须聚合185个快照跨越两天才能获得一个可控的静态网络。 S_t 和 S_s 之间的差异在所有系统中都显著, 对于面对面互动获得 $P < 0.001$, 蚂蚁为 $P < 0.05$, 蛋白质网络为 $P < 0.001$, 技术网络为 $P < 0.001$ 。

为了确定这些可控性提升在多大程度上取决于底层快照的拓扑结构, 我们使用随机化版本的实证数据计算 S_t 和 S_s 。

null模型会打乱整体网络结构、边的相对时间顺序, 或两者 (19)。在所有情况下, 我们发现 $S_t < S_s$ (图S8), 表明观察到的优势与网络随时间如何变化无关。我们还发现, 随着快照数量的增加, 时序网络比其静态对应物有更多机会达到完全可控性 (图S9), 这一发现通过使用考虑随机切换序列的玩具模型进一步验证 (SM部分S4)。最后, 我们还探讨了可控性的增益是否取决于节点之间交互的时间, 特别是事件间特征。为此, 我们随机分配每个交互事件的时间戳 (图S7), 抹去了数据中的固有突发性 (活跃期与相对静止期交替出现的时期) 和其他时间相关性 (28), 同时保持每个节点涉及的交互数量不变 $\langle k \rangle < \langle k \rangle$ (19), SM部分S₅。我们仍然发现, 在所有情况下 $S_t < S_s$ (图S13), 表明观察到的可控性增益与时间信号的特性无关。综合结果, 通过使用真实和合成数据获得, 表明时序性本身可以显著提高可控性。

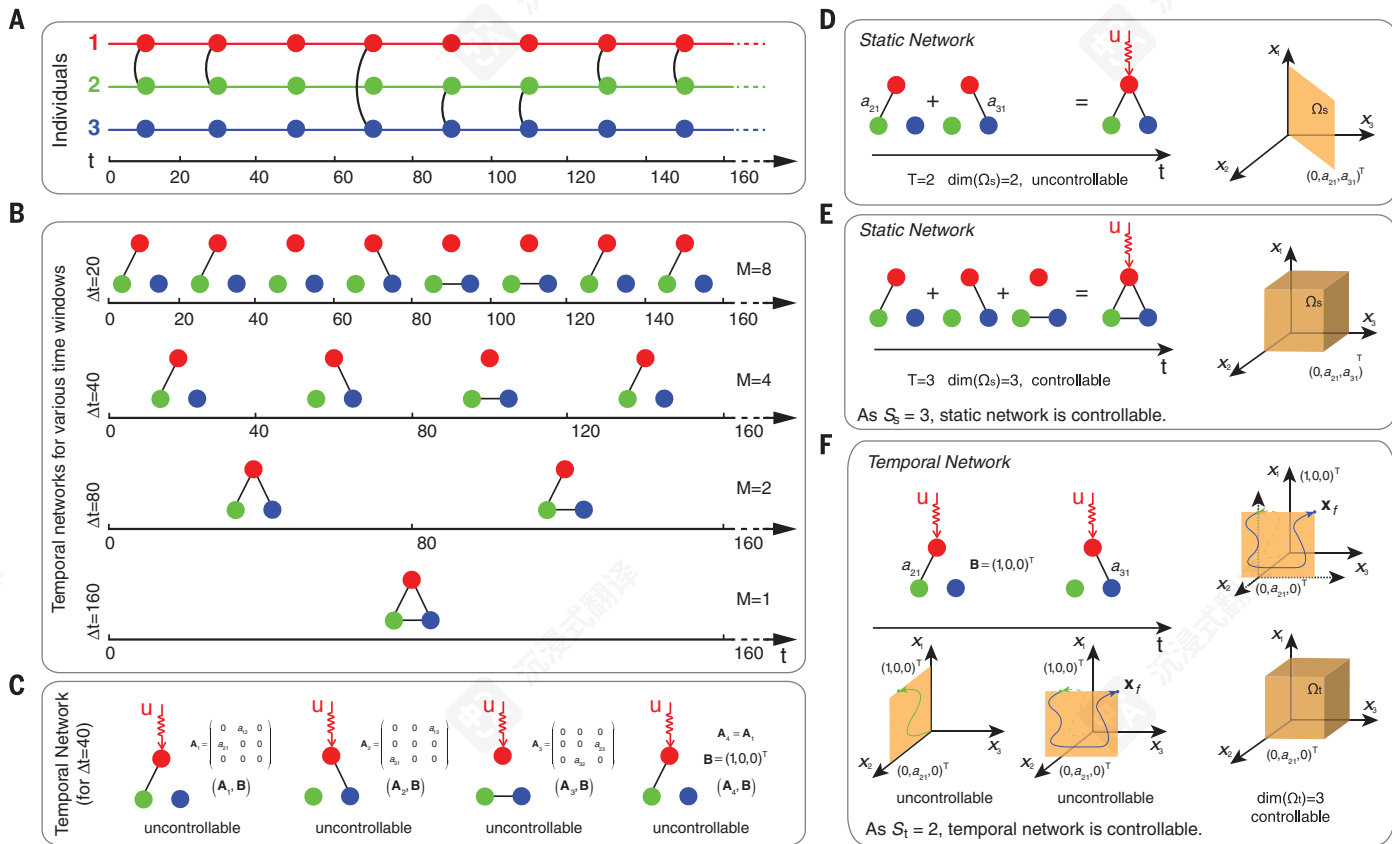


Fig. 1. Temporal versus static networks. (A) The sequence of contacts among three nodes, capturing, for example, the connection patterns between three individuals. We draw a line between two nodes if they interact with each other during a 20 s interval. (B) The temporal networks constructed from the contact sequence shown in (A) depend on the length of the time window Δt over which we aggregate the interactions. For short Δt , we obtain a large number of disconnected networks (snapshots); for sufficiently large Δt , these collapse into a single static network. (C) Four snapshots of a temporal network, constructed from (A) for $\Delta t = 40$. According to Kalman's rank condition (8), none of the snapshots are individually controllable from the top node. Yet, Eq. 3 predicts that the temporal network becomes controllable after the second snapshot. (D) A static network, obtained by aggregating the first two

snapshots in (C), is uncontrollable from the top node, as the controllable space Ω_s is two dimensional. This is illustrated by the yellow region on the right showing the set of all points $\mathbf{x}_f$ that can be reached in finite time from a given initial state $\mathbf{0}$ with an appropriate $\mathbf{u}(t)$. (E) We must aggregate at least three ($S_s = 3$) for the corresponding static network to become controllable. (F) The first two snapshots of a temporal network (top) shown in (C) and the controllable space for each snapshot (bottom). In this case, the temporal network becomes controllable after $S_t = 2$ snapshots. In other words, an appropriate signal acting on the green node can send the system to an arbitrary point in the $x_1 - x_2$ plane; a subsequent temporal signal acting on the blue node can take the system to the desired x_3 point in the space. Consequently, the controllable space of the temporal network is the full cube in the bottom right corner.

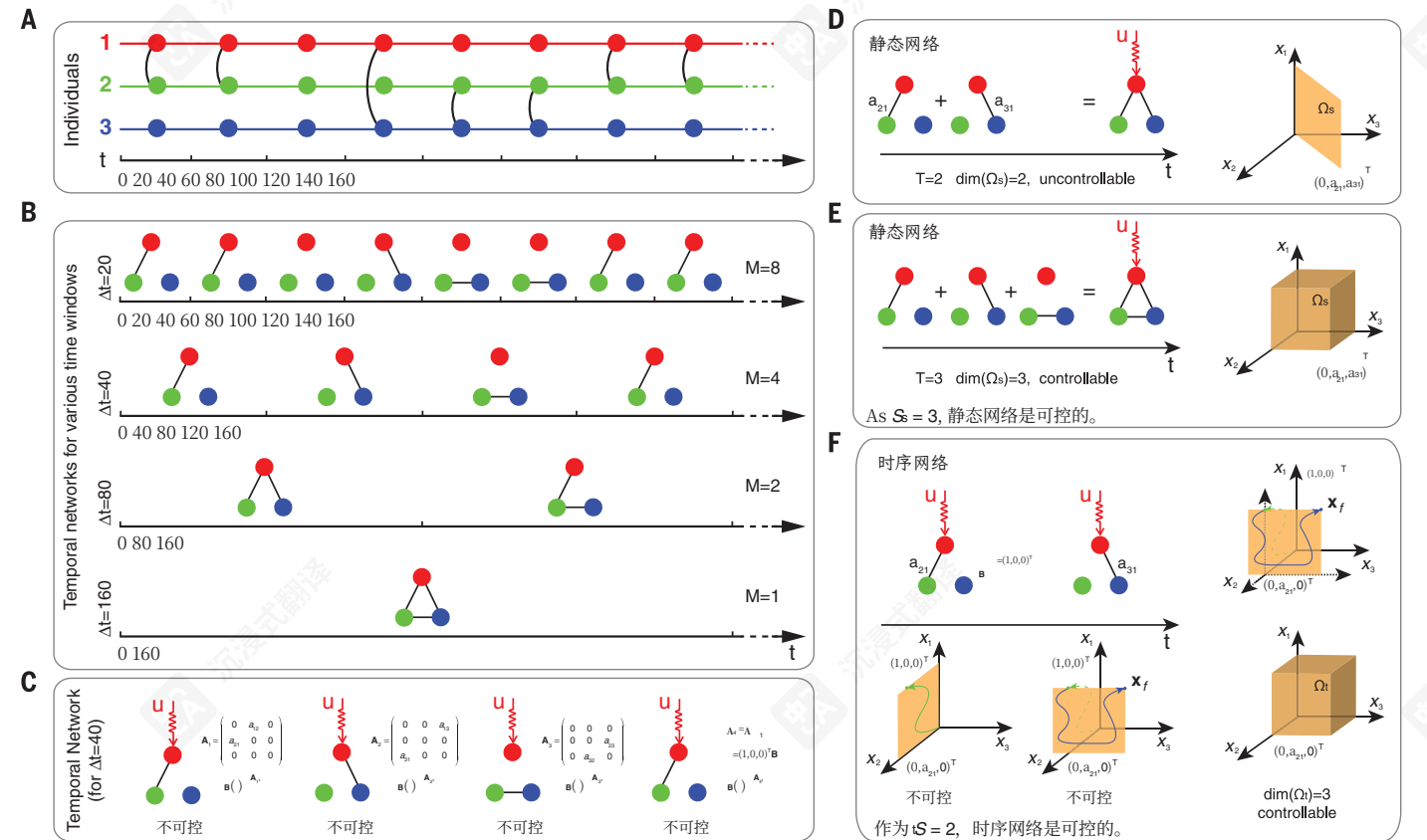


图1. 时序网络与静态网络。 (A) 三个节点之间的接触序列, 例如, 捕捉三个人之间的连接模式。如果两个节点在20秒的间隔内相互作用, 我们就在这两个节点之间画一条线。 (B) 从 (A) 中显示的接触序列构建的时序网络取决于我们聚合交互的时间窗口长度 Δt 。对于较短的 Δt , 我们得到大量不连接的网络 (快照); 对于足够大的 Δt , 这些合并成一个静态网络。 (C) 一个时序网络的四个快照, 根据卡尔曼秩条件 (8), 这些快照中没有一个是顶部节点单独可控的。然而, 方程式3预测, 时序网络在第二个快照之后变得可控。 (D) 通过聚合 (C) 中的前两个快照得到的静态网络, 从顶部节点不可控, 因为可控空间 $\Omega_s = 40$ 是二维的。这通过右侧的黄色区域说明, 该区域显示了从给定初始状态 $\mathbf{0}$ 和适当的 $\mathbf{u}(t)$ 在有限时间内可以到达的所有点 $\mathbf{x}_f$ 的集合。 (E) 我们必须聚合至少三个 ($S_s = 3$), 以便相应的静态网络变得可控。 (F) (C) 中显示的时序网络的前两个快照 (顶部) 和每个快照的可控空间 (底部)。在这种情况下, 时序网络在 $S_t = 2$ 个快照之后变得可控。换句话说, 作用于绿色节点的适当信号可以将系统发送到 $x_1 - x_2$ 平面中的任意点; 随后的作用于蓝色节点的时序信号可以将系统带到空间中所需的 x_3 点。因此, 时序网络的可控空间是右下角的全立方体。

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interactions involving each node unchanged [(19), SM section S4]. We still find that $S_t < S_s$ in all cases (fig. S13), indicating that the observed controllability gains are independent of the nature of temporal signal. The combined results, obtained by using real and synthetic data, indicate that temporality alone can considerably improve controllability.

It is well established that sparse and/or heterogeneous networks—common among real networks—face barriers to control, requiring many more driver nodes than dense or homogeneous networks. Our results indicate that natural and human-built systems can overcome this barrier by evolving or being engineered to be temporal. For example, systems like the Association for Computing Machinery (ACM) conference and the technological network, which show dramatic differences between S_s and S_t , can be steered to a large number of states within a short time window. Deviations from the strong link we observe between temporality and controllability are also informative. For instance, the slower growth of the controllable space and correspondingly modest gains in global controllability for the temporal ant-to-ant and protein-protein networks imply that the topology of these systems changes in ways that tend to revisit previous network structures and dynamics. As such, temporality, though present, offers less flexibility and may point to the importance of partial or “target” control in these systems (29). Taken together, we may be able to use the difference between S_s and S_t as a measure of the flexibility offered by temporality for a given system.

Our ability to control a system is determined not only by the time it takes to reach controllability but also by the amount of effort (energy) required to reach a particular final state. We develop a formalism to calculate the minimum input energy $\frac{1}{2} \int_{t_0}^T \mathbf{u}^T(t) \mathbf{u}(t) dt$ required to drive a

temporal network from an initial state $\mathbf{x}_0$ to final state $\mathbf{x}_f$. For systems following (Eq. 1), we have (SM section S5)

$$E(\mathbf{x}_0, \mathbf{x}_f) = \frac{1}{2} \mathbf{d}^T \mathbf{W}_{\text{eff}}^{-1} \mathbf{d} \quad (4)$$

where $\mathbf{d}$ is the difference vector between the desired final state $\mathbf{x}_f$ and the natural final state that the system reaches without control inputs, and the $N \times N$ matrix $\mathbf{W}_{\text{eff}}$ encodes the energy structure of the temporal network. For identical snapshots, $\mathbf{W}_{\text{eff}}$ reduces to the controllability matrix and Eq. 4 provides the control energy of a static network (SM, section S5.4). Computing $\mathbf{W}_{\text{eff}}$ scales in time as $\mathcal{O}(MN^4N_d)$, where N_d is the number of driver nodes, and involves $\mathcal{O}(M)$ nearly noninvertible matrices, with orders-of-magnitude differences in their elements. Hence, the limiting factor in calculating (Eq. 4) are the number of nodes and snapshots, not the number of interactions. This places high demands on numerical precision and renders the control energy calculation challenging for large networks. We can, however, numerically determine the energy for the technological system ($N = 34$, $M = 2$), but not for the next larger system, the protein-interaction network ($N = 74$, $M = 33$). We therefore calculate the minimum control energy E for small synthetic networks and test the implications of our results in real systems using the technological network.

Figure 3 compares the control energy (Eq. 4) of a synthetic temporal network and the technological network with that of their static counterparts as a function of Δt . We find that the average control energy of a temporal network is many orders of magnitude smaller than that of the corresponding static network, particularly in the highly temporal regime (with very small Δt). Indeed, for $\Delta t = 10^{-6}$, the energy difference between the static and temporal network exceeds 130

orders of magnitude (Fig. 3A). In other words, high temporality, corresponding to rapid changes in the network topology, offers remarkable energy savings. Applying this to a real system, we find that the energy required to control the technological network of Fig. 2D decreases by 274 orders of magnitude when the true temporal nature of the network is taken into account (Fig. 3B). In both temporal and static networks, we can reduce the control energy by using more driver nodes (24) (fig. S15). Yet the gap between the temporal and static network persists until all nodes are directly controlled (fig. S15). Finally, a direct comparison of the control energy distributions for static versus temporal networks reveals only a small overlap in certain regimes (Fig. 3, insets). This indicates that the worst-case control direction in a temporal network is often better than the best-case control direction in its static counterpart.

In some systems, from social systems to biological ones, a full comparison of the control cost of temporal and static networks might require us to consider the costs necessary to switch the network structure in temporal networks, or conversely, the potential costs of maintaining the network structure in static networks. For example, it would take exceptional effort to force a regulatory network not to change in time. Lacking measurements and models to estimate these costs, we have neglected them here. Yet, given the truly exceptional control cost differences between static and temporal networks (Fig. 3), we note that these coordination costs would need to differ in many orders of magnitude to alter our fundamental conclusion that temporal networks typically require less control energy.

The extreme energy savings characterizing temporal networks arise because of the orders-of-magnitude difference in the energy required to move in different directions in the state space

交互中涉及每个节点的数量保持不变 (19), SM部分S4]. 我们仍然发现, 在所有情况下 $S_t < S_s$ (图S13), 表明观察到的可控性增益与时间信号的特性无关。综合结果, 通过使用真实和合成数据获得, 表明时序性本身可以显著提高可控性。

众所周知, 稀疏和/或异构网络——这些是真实网络中常见的类型——面临着控制障碍, 需要比密集或同构网络多得多的驱动节点。我们的结果表明, 自然和人造系统可以通过进化或被设计成具有时间性来克服这一障碍。例如, 像计算机协会 (ACM) 会议和技术网络这样的系统, 在 S_s 和 S_t 之间表现出巨大差异, 可以在短时间内引导到大量状态。我们观察到的时序性与可控性之间的强关联的偏差也具有信息量。例如, 时序反蚂蚁-反蚂蚁网络和蛋白质-蛋白质网络的可控空间增长较慢, 以及相应地全局可控性提升有限, 这意味着这些系统的拓扑结构发生变化, 倾向于重新出现先前的网络结构和动态。因此, 尽管时序性存在, 但它提供的灵活性较小, 可能表明在这些系统中部分或“目标”控制的重要性 (29)。总而言之, 我们或许可以使用 S_s 和 S_t 之间的差异, 作为衡量特定系统时序性所提供灵活性的指标。

我们控制一个系统的能力不仅取决于达到可控状态所需的时间, 还取决于达到特定最终状态所需的努力 (能量)。我们开发了一种形式化方法来计算将时序网络从初始状态 $\mathbf{x}$ 推动到最终状态 $\mathbf{x}$ 所需的最小输入能量 $\frac{1}{2} \int_{t_0}^T \mathbf{u}^T(t) \mathbf{u}(t) dt$ 。

对于遵循 (公式1) 的系统, 我们有 (SM 第 S5 节)

$$E(\mathbf{x}_0, \mathbf{x}_f) = \frac{1}{2} \mathbf{d}^T \mathbf{W}_{\text{eff}}^{-1} \mathbf{d} \quad (4)$$

其中 $\mathbf{d}$ 是期望最终状态 $\mathbf{x}_f$ 与系统在无控制输入时达到的自然最终状态之间的差分向量, 而 $N \times N$ 矩阵 $\mathbf{W}_{\text{eff}}$ 编码了时序网络的能量结构。对于相同的快照, $\mathbf{W}_{\text{eff}}$ 简化为可控性矩阵, 且等式 4 给出了静态网络 (SM, 第 S5.4 节) 的控制能量。计算 $\mathbf{W}_{\text{eff}}$ 的时间复杂度为 $\mathcal{O}(MN^4N_d)$, 其中 N_d 是驱动节点的数量, 并且涉及 $\mathcal{O}(M)$ 近似不可逆的矩阵, 其元素存在数量级的差异。因此, 计算 (等式 4) 的瓶颈在于节点数量和快照数量, 而非交互数量。这给数值精度提出了极高要求, 使得对于大型网络的控制能量计算变得具有挑战性。然而, 我们可以数值确定技术系统 ($N = 34, M = 2$) 的能量, 但无法确定下一个更大的系统——蛋白质相互作用网络 ($N = 74, M = 33$)。因此, 我们计算小型合成网络的最低控制能量 E , 并使用技术网络测试结果在真实系统中的影响。

图3比较了合成时序网络和技术网络的控制能量 (公式4) 与其静态对应物的控制能量随 Δt 的变化。我们发现, 时序网络的平均控制能量比相应的静态网络小很多数量级, 特别是在高度时序的范围内 (Δt 非常小)。实际上, 对于 $\Delta t = 10^{-6}$, 静态网络和时序网络之间的能量差超过130

时序网络和静态网络的控制能量分布的直接比较表明, 在某些范围内只有很小的重叠 (图3, 插图)。这表明, 时序网络的最坏情况控制方向通常比其静态对应物的最佳情况控制方向更好。

在一些系统中, 从社会系统到生物系统, 对时序网络和静态网络的控制成本进行全面比较, 可能需要我们考虑时序网络中切换网络结构的成本, 或者反之, 静态网络中维持网络结构的潜在成本。例如, 要强迫一个调控网络在时间上不发生变化, 需要付出极大的努力。由于缺乏估计这些成本的措施和模型, 我们在本文中忽略了它们。然而, 鉴于静态网络和时序网络之间确实异常的控制成本差异 (图3), 我们注意到, 这些协调成本需要相差几个数量级, 才能改变我们关于时序网络通常需要较少控制能量的基本结论。

时序网络所具有的极大幅度节能特性, 源于在状态空间中沿不同方向移动所需能量的巨大差异

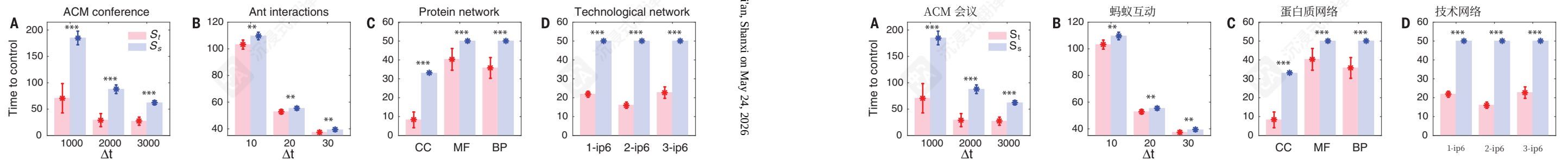


Fig. 2. Faster paths to controllability in temporal networks. Time needed to control four kinds of temporal networks: (A) 20,818 face-to-face interactions between the attendees of an ACM hypertext conference with 113 participants, recorded over 2.5 days in 2009 (25); (B) 1911 antenna-body interactions between 89 ants over 1438 s (26); (C) dynamic protein-protein interactions in the yeast *Saccharomyces cerevisiae* annotated by the three domains of gene ontology—cellular component (CC), molecular function (MF), and biological process (BP). The data consist of 22,570 protein-protein interactions between 84, 74, and 85 nodes, recorded within 33, 50, and 50 snapshots, respectively (SM section S2) (27); (D) data packet exchanges in an emulated mobile ad hoc network, capturing 304, 383, and 3287 interactions between 34 nodes (mobile devices) occurring

over 50 snapshots each from three simulations, 1-ip6, 2-ip6, and 3-ip6 (SM section S2). For each time window Δt , S_t is the minimum number of snapshots required to achieve full control of a temporal network, and S_s is the minimum number of snapshots we must aggregate to obtain a controllable static network. We find that $S_t < S_s$ for any Δt , both for the original sequence of snapshots and when the sequence of interactions is randomized, meaning that the result is not an artifact of the original data (see SM sections S3 and S4). For the significance test, we obtain $***P < 0.001$ and $**P < 0.05$; Student's t test. The weight of each link is set randomly between 0 and 1. The number of driver nodes is fixed to 20% of the network size, and each point represents an average over 10^3 realizations of link weights on the same network structure (temporal or static).

图 2. 时序网络中更快达到可控性的路径。控制四种时序网络所需的时间: (A) 2009 年 2.5 天内记录的 113 名参与者 ACM 超文本会议与会者之间的 20,818 次面对面互动 (25); (B) 89 只蚂蚁在 1438 秒内进行的 1911 次天线-身体互动 (26); (C) 酵母酿酒酵母中动态的蛋白质-蛋白质互动, 由基因本体学的三个域标注——细胞组分 (CC)、分子功能 (MF) 和生物学过程 (BP)。数据包括 84、74 和 85 个节点之间 22,570 次蛋白质-蛋白质互动, 分别记录在 33、50 和 50 个快照内 (SM 部分 S2) (27); (D) 模拟移动自组网中的数据包交换, 捕获了 34 个节点 (移动设备) 之间发生的 304、383 和 3287 次互动

来自三个模拟 (1-ip6、2-ip6 和 3-ip6), 每个模拟超过 50 个快照 (SM 第 S2 节)。对于每个时间窗口 Δt , S_t 是实现时序网络完全控制所需的最少快照数, 而 S_s 是我们必须聚合以获得可控静态网络的最少快照数。我们发现, 对于任何 Δt , $S_t < S_s$ 在原始快照序列中以及当交互序列随机化时都成立, 这意味着结果并非原始数据的假象 (参见 SM 第 S3 和 S4 节)。对于显著性检验, 我们获得 $***P < 0.001$ 和 $**P < 0.05$; 学生 t 检验。每条边的权重在 0 和 1 之间随机设置。驱动节点的数量固定为网络规模的 20%, 每个点代表在相同网络结构 (时序或静态) 上对边权重 10^3 次实现取的平均值。

(24). In a static network, if we must travel in an energetically costly direction, we have no choice but to spend the required energy to “push” against the system’s dynamics. By contrast, in a temporal network, we can exploit the changing topology to avoid these expensive directions. Much like navi-

gating a sailboat, it is easier to travel in a particular direction if we exploit the shifts in the wind direction (the vector field of the network dynamics): We raise the sails when the wind helps us and pull them back when it works against us. In other words, we only push toward the desired

final state when the topology renders the energy cost acceptable, and pause when the topology makes the cost prohibitive.

The orders-of-magnitude decrease in control energy induced by temporality has implications for natural and human-built systems alike. Our calculations (Fig. 3) show that the key determinant of a temporal system’s energy advantage is the parameter Δt representing how long each snapshot is active. Systems in which the network’s structure evolves rapidly (relative to its intrinsic dynamics) can have huge savings in terms of the effort needed to control them. By contrast, if the network’s structure evolves slowly relative to its dynamics, then static and temporal networks require comparable control energy. Indeed, as shown in Fig. 3, for small Δt , temporal networks display a decided energy advantage; by contrast, for large Δt , the energy advantage of temporal networks is not obvious. The exact Δt value at which transition occurs depends on the number of driver nodes (N_d), as well as the details of the network’s structure such as its average degree and link strengths.

Real systems often obey constraints that forbid the states of the nodes from taking arbitrary values. For instance, the generator frequencies in the power grid can only vary within a narrow range around their normal operating point, without inducing failures, and metabolite concentrations (I) within a cell must always be nonnegative. These limitations mean that the control trajectories cannot wander arbitrarily far into the state space but must exhibit a high degree of locality (23).

To test the degree of locality in temporal networks, we calculate the length of each control trajectory using $L = \int_0^t \|\dot{\mathbf{x}}(t)\| dt$, where the energy-optimal trajectory for temporal networks as they move from $\mathbf{x}_0$ to $\mathbf{x}_f$ follows

$$\mathbf{x}(t) = e^{\mathbf{A}_m(t-t_{m-1})}\mathbf{x}(t_{m-1}) + \mathbf{W}_m[t_{m-1}, t]\mathbf{c}_m^*$$

for $t \in [t_{m-1}, t_{m-1} + \tau_m]$, where $\mathbf{W}_m[t_{m-1}, t]$ is the matrix corresponding to the snapshot m and $\mathbf{c}_m^*$ is a constant vector of dimension N (SM section S9). In general, L depends on the initial state $\mathbf{x}_0$ and the state-space distance $\delta = \|\mathbf{x}_f - \mathbf{x}_0\|$ between the origin and destination points (Fig. 4A, A and B). For $\mathbf{x}_0 = \mathbf{0}$, in both static and temporal networks, L increases linearly with δ (Fig. 4B). Yet for any δ , the optimal control trajectories in temporal networks are about five orders of magnitude shorter than trajectories in their static counterparts (Fig. 4B). The difference is particularly notable in real systems: For the technological network, the temporal trajectory is 29 orders of magnitude shorter than the trajectory for the corresponding static network (Fig. 4C). Indeed, it is known that, in static networks, L can be large even when δ approaches zero, implying static control generally demands highly nonlocal control trajectories (23). By contrast, we find that the dynamical flexibility offered by temporality allows $\mathbf{x}(t)$ not to have to wander far into phase space as it moves from $\mathbf{x}_0$ to $\mathbf{x}_f$. The sailing analogy helps us once again to understand this

(24). 在静态网络中，如果我们必须沿能量消耗大的方向移动，就别无选择地需要花费所需能量来“对抗”系统的动态变化。相比之下，在时序网络中，我们可以利用拓扑结构的变化来避开这些高成本方向。这就像导航时……

操控帆船时，如果我们利用风向的变化（即网络动力学的矢量场），朝特定方向航行会更轻松：风助我们时扬帆，风逆我们时收帆。换句话说，我们只朝期望的方向推进，当拓扑结构使能量成本可接受时，才推向最终状态，当拓扑结构使成本过高时，则暂停。

最终状态，当拓扑结构使能量成本可接受时，暂停，当拓扑结构使成本过高时。

时序网络带来的控制能量数量级下降，对自然系统和人类构建的系统都有影响。我们的计算（图3）表明，时序系统能量优势的关键决定因素是参数 Dt ，它代表每个快照的活跃时长。网络结构快速演化（相对于其内在动力学）的系统，在控制所需付出的努力上能节省巨大。相比之下，如果网络结构演化缓慢（相对于其动力学），静态网络与时序网络的控制能量相当。实际上，如图3所示，对于较小的 Dt ，时序网络显示出明显的能量优势；而对于较大的 Dt ，时序网络的能量优势并不明显。转换发生的精确 Dt 值取决于驱动节点数量（ d ）以及网络结构的细节，如平均度数和连接强度。

实际系统通常遵循限制条件，禁止节点的状态取任意值。例如，电网中的发电机频率只能在正常工作点附近一个狭窄的范围内变化，而不会引发故障，细胞内的代谢物浓度（ I ）必须始终为非负值。这些限制意味着控制轨迹不能随意地深入状态空间，而必须表现出高度的局部性（23）。

为了测试时序网络中的局部性程度，我们使用 $L = \int_0^t \|\dot{\mathbf{x}}(t)\| dt$ 计算每个控制轨迹的长度，其中时序网络从 $\mathbf{x}_0$ 移动到 $\mathbf{x}_f$ 的能量最优轨迹遵循

$$\mathbf{x}(t) = e^{\mathbf{A}_m(t-t_{m-1})}\mathbf{x}(t_{m-1}) + \mathbf{W}_m[t_{m-1}, t]\mathbf{c}_m^*$$

for $t \in [t_{m-1}, t_{m-1} + \tau_m]$, 其中 $\mathbf{W}_m[t_{m-1}, t]$ 是与快照 m 对应的矩阵， $\mathbf{c}_m^*$ 是维度为 N 的常量向量（SM 节 S9）。通常， L 取决于初始状态 $\mathbf{x}_0$ 以及起点和终点之间的状态空间距离

$\delta = \|\mathbf{x}_f - \mathbf{x}_0\|$ （图4，A和B）。对于 $\mathbf{x}_0 = \mathbf{0}$ ，在静态网络和时序网络中， L 都随 δ 线性增加（图4B）。然而，对于任何 δ ，时序网络中的最优控制轨迹都比其静态对应物的轨迹短约五个数量级（图4B）。这种差异在实际系统中尤为明显：对于技术网络，时序轨迹比对应静态网络的轨迹短29个数量级（图4C）。事实上，众所周知，即使在 δ 接近零时，静态网络中的 L 也可能很大，这意味着静态控制通常需要高度非局部的控制轨迹（23）。相比之下，我们发现时序性提供的动态灵活性允许 $\mathbf{x}(t)$ 在从 $\mathbf{x}_0$ 移动到 $\mathbf{x}_f$ 时不必深入相空间很远。航海类比再次帮助我们理解这一点

Fig. 3. Temporal networks require less control energy. (A) Control energy for temporal and static networks. We consider a temporal network with two snapshots, each lasting Δt time. $\langle E \rangle$ (solid line) is the average minimum energy over 10^4 randomly selected final states $\mathbf{x}_f$ from a unit sphere centered on $\mathbf{x}_0 = \mathbf{0}$. The shaded area is enclosed by the minimum and maximum E obtained numerically. $\bar{E}$ (dashed line) is the theoretical upper bound of E (SM section S6). With increasing Δt , $\bar{E}$ (and $\langle E \rangle$) decreases as $\bar{E} \sim \Delta t^{-\gamma}$, where γ is indicated with a black dashed line, before reaching a plateau. The fraction of driver nodes is 0.1, and increasing the number of driver nodes reduces the necessary control energy both for temporal and static networks, yet temporal networks continue to require less energy (see fig. S15 for the case of more driver nodes). Insets show the distribution of minimal energy for two specific Δt . The small overlap between temporal and static networks indicates that temporal networks are energetically preferable and that this is largely independent of the control direction (fig. S15). Here, $a_1 = -3$, $a_2 = -1$, $N = 20$. a_1 and a_2 are self-loop weights chosen to stabilize the standalone dynamics of each snapshot, and the average degree within each snapshot is 6 (see SM section S8 for details). (B) The distribution of E with one driver node for the technological network with $\Delta t = 10^{-6}$ (see fig. S15 for more details).

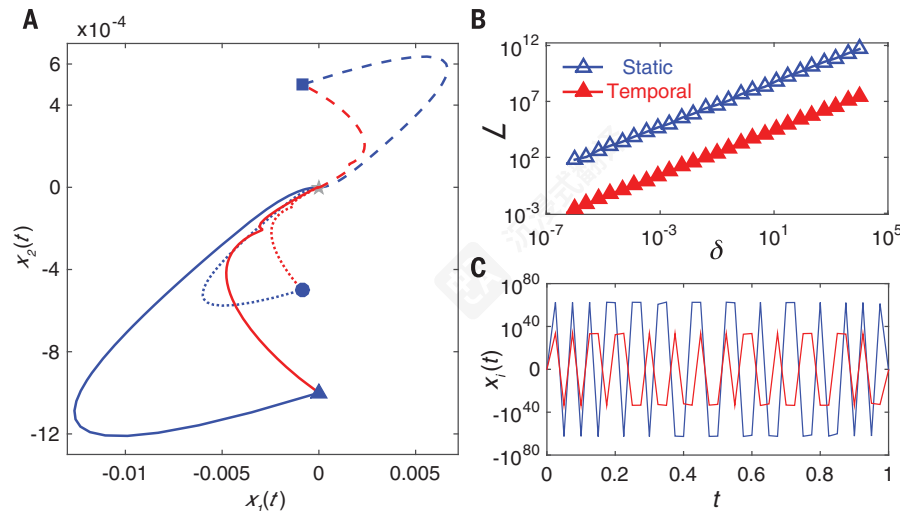
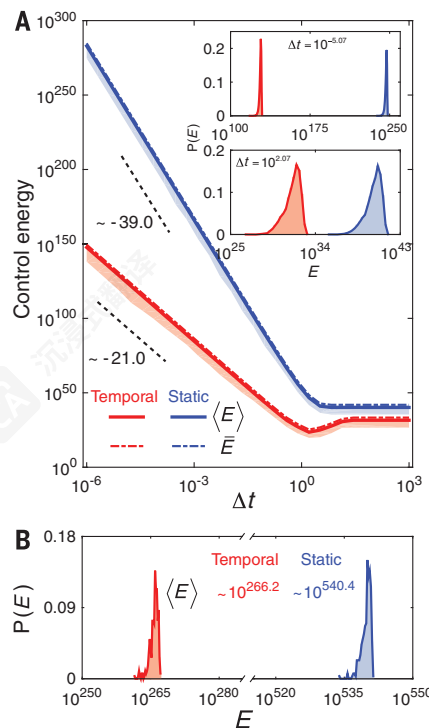


Fig. 4. Temporal networks exhibit more local trajectories. (A) Three trajectories (triangle, circle, and square) for static (blue lines) and temporal (red lines) networks starting from $\mathbf{x}_0 = \mathbf{0}$, controlled to reach $\|\mathbf{x}_f\| = 10^{-3}$ after a unit time with $a_1 = -10$ and $a_2 = 2$. (B) The length of the control trajectory L as a function of control distance $\delta = \|\mathbf{x}_f - \mathbf{x}_0\|$. L is always much smaller for the temporal networks than for their static counterparts, regardless of the control distance (see SM section S9). Each point represents an average over 10^4 final states; we choose $N = 10$, $M = 5$. See SM section S9 for other parameters. (C) For the technological network, we track the evolution of $x_1(t)$ (corresponding to the maximum length of all state components) from $x_1(0) = 0$ to $x_1(1) = 1/\sqrt{34}$, finding that L is in the order of 10^{35} for the temporal network, in contrast with 10^{64} for the corresponding static network (see SM section S9 for more details).

图3. 时序网络需要的控制能耗更低。(A) 时序网络和静态网络的控制能耗。我们考虑一个包含两个快照的时序网络，每个快照持续 Dt 时间。 $\langle E \rangle$ (实线) 表示在以 $\mathbf{x}_0 = \mathbf{0}$ 为中心的单位球面上随机选择的最终状态 $\mathbf{x}_f$ 的平均最小能耗。阴影区域由数值计算得到的 E 的最小值和最大值围成。 $\bar{E}$ (虚线) 是 SM 第 S6 节中 E 的理论上限。随着 Dt 的增加， $\bar{E}$ (和 $\langle E \rangle$) 按 $\bar{E} \sim Dt^{-\gamma}$ 的规律下降，其中 γ 用黑色虚线表示，直到达到平台期。驱动节点的比例为 0.1，增加驱动节点的数量可以降低时序网络和静态网络所需的控制能耗，但时序网络仍然需要更少的能量（参见 fig. S15 中驱动节点数量更多的情况）。插图显示了两个特定 Dt 的最小能耗分布。时序网络和静态网络之间的小重叠表明时序网络在能耗上更优，并且这与控制方向基本无关（fig. S15）。这里， $a_1 = -3$ ， $a_2 = -1$ ， $N = 20$ 。 a_1 和 a_2 是自环权重，用于稳定每个快照的独立动力学，每个快照内的平均度为 6（详情见 SM 第 S8 节）。(B) 技术网络在 $Dt = 10^{-6}$ 时 E 的分布（更多细节见图 S）。

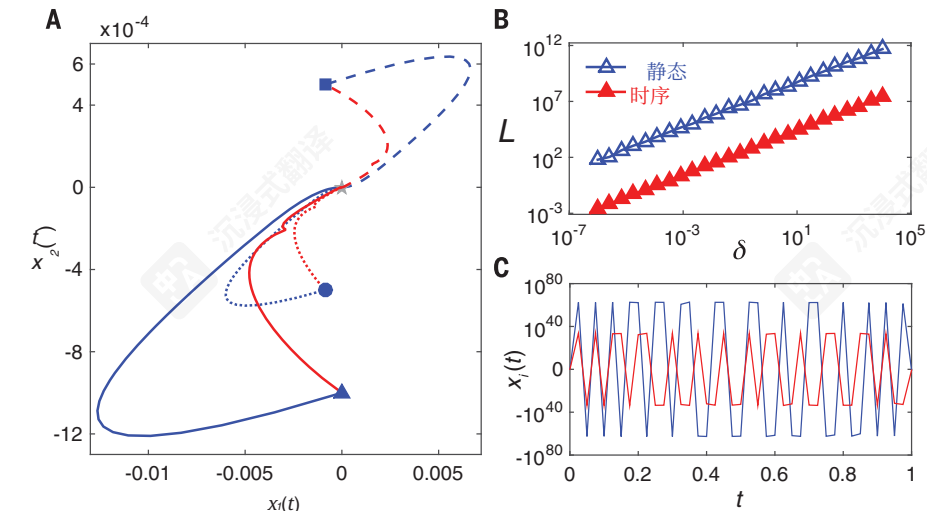
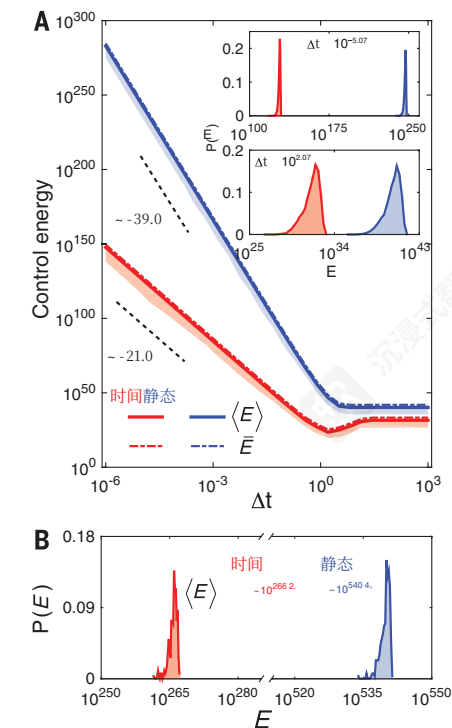


图4. 时序网络表现出更多局部轨迹。(A) 静态（蓝色线）和时序（红色线）网络从 $\mathbf{x}_0 = \mathbf{0}$ 开始的三个轨迹（三角形、圆形和方形），控制以在单位时间内到达 $\|\mathbf{x}_f\| = 10^{-3}$ ，控制距离为 $a_1 = -10$ 和 $a_2 = 2$ 。(B) 控制轨迹长度 L 作为控制距离 $d = \|\mathbf{x}_f - \mathbf{x}_0\|$ 的函数。时序网络的 L 始终远小于其静态对应物的 L ，无论控制距离如何（参见 SM 第 S9 节）。每个点代表 10^4 个最终状态的平均值；我们选择 $N = 10$ ， $M = 5$ 。其他参数参见 SM 第 S9 节。(C) 对于技术网络，我们跟踪 $x_1(t)$ （对应于所有状态组件的最大长度）从 $x_1(0) = 0$ 到 $x_1(1) = 1/\sqrt{34}$ 的演变，发现时序网络的 L 为 10^{35} 级，而对应静态网络的 L 为 10^{64} （更多细节参见 SM 第 S9 节）。

difference: Travel against the prevailing wind is possible, but we must use complicated zigzag maneuvers to reach our destination. If, however, the wind occasionally changes direction, a strategic use of the sails allows us to travel more directly to our destination.

The stark drop in control trajectory lengths observed here helps us resolve a fundamental conundrum about the inner workings of real systems. In real systems (particularly in sub-cellular networks), only a small number of “sensory” nodes are generally available for direct manipulation; however, the use of fewer driver nodes can drastically increase the length of the control trajectories (23), meaning that the nodes will have to take up extreme state values as the system converges to the desired final state. Temporality unlocks a mechanism through which typically underactuated natural systems can smoothly glide between required states—e.g., different stages of the cell cycle—without the need to pass through highly extreme states that might be potentially destructive to the system as a whole. That is, we can drive these sytems to their desired final state without drastically altering the state variables of the individual nodes. This could be particularly important in subcellular networks, where large changes in the concentrations of individual molecules may be toxic for a cell, or in technological systems, where the individual components must operate within their design specifications.

Our results demonstrate that temporality has profound dynamical consequences, fundamentally improving our ability to control real networks. This is consistent with previous findings that nonlinear dynamics, in this case induced by temporality, can be an asset rather than an obstacle to control (30). Beyond control, the measures introduced here offer a general glimpse into

the dynamical flexibility of temporal networks. Indeed, these networks that exhibit a rapid growth of their controllable space (small S_c), and with concomitantly low control costs, represent dynamical “Swiss army knives,” capable of displaying a wide range of behaviors with a relatively small number of network links. Thus, our findings open the door to the design of networks that are adaptive to changing environments while using the same set of underlying nodes. This insight may lead to new, robust cyber networks, adaptive “smart” infrastructures, and new strategies for intervening in natural systems that capitalize on the best parts of each snapshot to obtain “the best of all worlds.”

REFERENCES AND NOTES

1. E. Almaas, B. Kovács, T. Vicsek, Z. N. Oltvai, A.-L. Barabási, *Nature* **427**, 839–843 (2004).
2. R. Cohen, S. Havlin, *Complex Networks: Structure, Robustness and Function* (Cambridge Univ. Press, 2010).
3. Z. Toroczkai, H. Guclu, *Physica A* **378**, 68–75 (2007).
4. N. Masuda, K. Klemm, V. M. Eguiluz, *Phys. Rev. Lett.* **111**, 188701 (2013).
5. H. H. K. Lentz, T. Selhorst, I. M. Sokolov, *Phys. Rev. Lett.* **110**, 118701 (2013).
6. I. Scholtes et al., *Nat. Commun.* **5**, 5024 (2014).
7. M. Starnini, A. Baronchelli, A. Barrat, R. Pastor-Satorras, *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* **85**, 056115 (2012).
8. R. E. Kalman, *J. Soc. Ind. Appl. Math. Ser. A* **1**, 152–192 (1963).
9. J. Klamka, *Controllability of Dynamical Systems* (Mathematics and Its Applications Series, Springer, 1991), vol. 48.
10. P. Ögren, E. Fiorelli, N. Leonard, *IEEE Trans. Automat. Contr.* **49**, 1292–1302 (2004).
11. N. Yosef, A. Regev, *Cell* **144**, 886–896 (2011).
12. J. Uhlendorf et al., *Proc. Natl. Acad. Sci. U.S.A.* **109**, 14271–14276 (2012).
13. T. Nepusz, T. Vicsek, *Nat. Phys.* **8**, 568–573 (2012).
14. G. Chen, *Int. J. Control. Autom. Syst.* **12**, 221–230 (2014).
15. Y.-Y. Liu, J.-J. Slotine, A.-L. Barabási, *Nature* **473**, 167–173 (2011).
16. I. Rajapakse, M. Groudine, M. Mesbahi, *Proc. Natl. Acad. Sci. U.S.A.* **108**, 17257–17262 (2011).
17. J. Ruths, D. Ruths, *Science* **343**, 1373–1376 (2014).
18. G. Chen, *Int. J. Autom. Comput.* **14**, 1–9 (2017).

19. P. Holme, J. Saramäki, *Phys. Rep.* **519**, 97–125 (2012).
20. M. Pósfai, P. Hövel, *New J. Phys.* **16**, 123055 (2014).
21. Y.-Y. Liu, A.-L. Barabási, *Rev. Mod. Phys.* **88**, 035006 (2016).
22. G. Xie, D. Zheng, L. Wang, *IEEE Trans. Automat. Contr.* **47**, 1401–1405 (2002).
23. J. Sun, A. E. Motter, *Phys. Rev. Lett.* **110**, 208701 (2013).
24. G. Yan et al., *Nat. Phys.* **11**, 779–786 (2015).
25. L. Isella et al., *J. Theor. Biol.* **271**, 166–180 (2011).
26. B. Blonder, A. Dornhaus, *PLOS ONE* **6**, e20298 (2011).
27. J. Wang, X. Peng, M. Li, Y. Pan, *Proteomics* **13**, 301–312 (2013).
28. A.-L. Barabási, *Nature* **435**, 207–211 (2005).
29. J. Gao, Y.-Y. Liu, R. M. D'Souza, A.-L. Barabási, *Nat. Commun.* **5**, 5415 (2014).
30. S. P. Cornelius, W. L. Kath, A. E. Motter, *Nat. Commun.* **4**, 1942 (2013).

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SUPPLEMENTARY MATERIALS

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Materials and Methods
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差异：逆风航行是可能的，但我们必须使用复杂的之字形机动才能到达目的地。然而，如果风向偶尔改变，策略性地使用帆可以让我们的航行更直接地到达目的地。

这里观察到的控制轨迹长度急剧下降，有助于我们解决关于真实系统内部运作的一个基本难题。在真实系统（尤其是在亚细胞网络中），通常只有少量“传感”节点可用于直接操控；然而，使用较少的驱动节点会显著增加控制轨迹的长度（23），这意味着节点必须在系统收敛到期望的最终状态时采取极端状态值。时序性解锁了一种机制，使通常欠驱动性的自然系统能够在无需通过可能对整个系统造成破坏的极端状态的情况下，平滑地在所需状态之间过渡——例如，细胞周期的不同阶段。也就是说，我们可以驱动这些系统达到其期望的最终状态，而无需大幅改变单个节点的状态变量。这在亚细胞网络中可能尤为重要，因为单个分子浓度的巨大变化可能对细胞具有毒性，或者在技术系统中，单个组件必须在设计规格范围内运行。

我们的结果表明时序性具有深刻的动力学后果，从根本上提升了我们控制真实网络的能力。这与先前发现相一致，即非线性动力学——在这种情况下由时序性诱导——可以成为控制的优势而非障碍（30）。除了控制之外，这里引入的度量为我们提供了对时序网络动力学灵活性的总体洞察。

事实上，这些表现出其可控空间（小 S_c ）快速增长的时序网络，以及随之而来的低控制成本，代表了动态的“瑞士军刀”，能够通过相对较少的网络连接展示广泛的行为。因此，我们的发现为设计能够适应变化环境且使用相同底层节点的网络打开了大门。这一见解可能引领新型、鲁棒的计算机网络、自适应“智能”基础设施，以及利用每个快照的最佳部分以获得“尽善尽美”的新策略来干预自然系统。

参考文献与注释

1. E. Almaas, B. Kovács, T. Vicsek, Z. N. Oltvai, A.-L. Barabási, *Nature* **427**, 839–843 (2004).
2. R. Cohen, S. Havlin, *Complex Networks: Structure, Robustness and Function* (Cambridge Univ. Press, 2010).
3. Z. Toroczkai, H. Guclu, *Physica A* **378**, 68–75 (2007).
4. N. Masuda, K. Klemm, V. M. Eguiluz, *Phys. Rev. Lett.* **111**, 188701 (2013).
5. H. H. K. Lentz, T. Selhorst, I. M. Sokolov, *Phys. Rev. Lett.* **110**, 118701 (2013).
6. I. Scholtes et al., *Nat. Commun.* **5**, 5024 (2014).
7. M. Starnini, A. Baronchelli, A. Barrat, R. Pastor-Satorras, *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* **85**, 056115 (2012).
8. R. E. Kalman, *J. Soc. Ind. Appl. Math. Ser. A* **1**, 152–192 (1963).
9. J. Klamka, *Controllability of Dynamical Systems* (Mathematics and Its Applications Series, Springer, 1991), vol. 48.
10. P. Ögren, E. Fiorelli, N. Leonard, *IEEE Trans. Automat. Contr.* **49**, 1292–1302 (2004).
11. N. Yosef, A. Regev, *Cell* **144**, 886–896 (2011).
12. J. Uhlendorf et al., *Proc. Natl. Acad. Sci. U.S.A.* **109**, 14271–14276 (2012).
13. T. Nepusz, T. Vicsek, *Nat. Phys.* **8**, 568–573 (2012).
14. G. Chen, *Int. J. Control. Autom. Syst.* **12**, 221–230 (2014).
15. Y.-Y. Liu, J.-J. Slotine, A.-L. Barabási, *Nature* **473**, 167–173 (2011).
16. I. Rajapakse, M. Groudine, M. Mesbahi, *Proc. Natl. Acad. Sci. U.S.A.* **108**, 17257–17262 (2011).
17. J. Ruths, D. Ruths, *Science* **343**, 1373–1376 (2014).
18. G. Chen, *Int. J. Autom. Comput.* **14**, 1–9 (2017).

19. P. Holme, J. Saramäki, *Phys. Rep.* **519**, 97–125 (2012).
20. M. Pósfai, P. Hövel, *New J. Phys.* **16**, 123055 (2014).
21. Y.-Y. Liu, A.-L. Barabási, *Rev. Mod. Phys.* **88**, 035006 (2016).
22. G. Xie, D. Zheng, L. Wang, *IEEE Trans. Automat. Contr.* **47**, 1401–1405 (2002).
23. J. Sun, A. E. Motter, *Phys. Rev. Lett.* **110**, 208701 (2013).
24. G. Yan et al., *Nat. Phys.* **11**, 779–786 (2015).
25. L. Isella et al., *J. Theor. Biol.* **271**, 166–180 (2011).
26. B. Blonder, A. Dornhaus, *PLOS ONE* **6**, e20298 (2011).
27. J. Wang, X. Peng, M. Li, Y. Pan, *Proteomics* **13**, 301–312 (2013).
28. A.-L. Barabási, *Nature* **435**, 207–211 (2005).
29. J. Gao, Y.-Y. Liu, R. M. D'Souza, A.-L. Barabási, *Nat. Commun.* **5**, 5415 (2014).
30. S. P. Cornelius, W. L. Kath, A. E. Motter, *Nat. Commun.* **4**, 1942 (2013).

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分析计算。A.L. 完成了所有数值计算（技术网络的分析除外，该部分由 Y.-Y.L. 独立完成）。A.-L.B.、A.L. 和 S.P.C. 领导了论文的撰写，Y.-Y.L. 和 L.W. 负责编辑论文。

补充材料

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The fundamental advantages of temporal networks

A. Li, S. P. Cornelius, Y.-Y. Liu, L. Wang, and A.-L. Barabási

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In network science, change is good

Historically, network science focused on static networks, in which nodes are connected by permanent links. However, in networked systems ranging from protein-protein interactions to social networks, links change. Although it might seem that permanent links would make it easier to control a system, Li *et al.* demonstrate that temporality has advantages in real and simulated networks. Temporal networks can be controlled more efficiently and require less energy than their static counterparts.

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时序网络的基本优势

A. Li, S. P. 科尔内利乌斯, Y.-Y. 刘, L. 王, 和 A.-L. 巴拉巴西

科学 358 (6366), DOI: 10.1126/science.aai7488

在网络科学中，变化是好的

历史上，网络科学专注于静态网络，其中节点通过永久链接相连。然而，在网络化系统中，从蛋白质-蛋白质相互作用到社交网络，链接都在变化。尽管永久链接似乎会使控制系统变得更容易，但李等人证明了时序性在真实和模拟网络中具有优势。时序网络比其静态对应物控制更高效，且所需能量更少。

科学，本期第1042页

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